

ENERGY CALIBRATION DETERMINATION FOR NaI(Tl) DETECTORS USING GAMMA RADIONUCLIDE SOURCES WITH HIGH BETA PLUS (β^+) BRANCHING RATIOS

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Abstract

For a proper detector calibration, the response of a scintillator detectors to gamma sources must directly depend on the energy that the gamma sources deposit. Adopting alternative methods to parameterize detector response functions within the relevant energy range is necessary when gamma sources are not readily available. The conversion of gamma source energies into voltage pulses can be simulated by computational methods by use of a sequence of random processes. To simulate the processes of photon and electron interaction processes within the cylindrical crystal as detected by a NaI(Tl) detector, an in-house Photon code (Calx) based on the Monte Carlo method was developed. A range of mono-energetic photon sources within the energies range of 0.2-2 MeV was applied to the scintillation detector and the simulated pulse height spectra were generated. The results revealed that the simulated spectrum of an idealized Na-22 gamma source was achieved to give the 1274 keV and 511 keV prominent peaks in the Na-22 spectrum. These new peaks based on significant β^+ peaks improved on the 3 other prominent simulated peaks obtained from the Cs-137 (662 keV) and Co-60 (1172 keV and 1332 keV) to get the best line of fit for the calibration curve. The results from this study were compared with the experimental results of Akkurt *et al.* and the deviation of FWHM for the simulated Na-22 source for energy 511 keV and 1274 keV where 8% and 7% respectively. The corresponding deviation in resolution energy for the two energies (1274 keV and 511 keV) was 7%. In conclusion, the added simulated Na-22 source modified the calibration equation and hence improved the NaI(Tl) detector calibration procedure used in low-level counting of radionuclides in samples analyzed in-house.

Keywords

: Monté Carlo simulation, Energy calibration, Branching ratio, Beta decay, Detector response

1. Background

A variety of radiation detectors are used for research in many different fields which are principally related to the detection of natural and artificial radiation [2]. In the study of gamma radiation, Thallium-Activated Sodium iodide (NaI(Tl)) detectors stand out of the pack of available detectors for its high gamma detection efficiency, cost and ease of use despite its poor resolution [3]. To perform gamma spectroscopy using NaI(Tl) detectors, detector response functions to gamma radiation are evaluated based on experimental conditions [4]. Calibrations techniques performed to determine these detector responses are specifically based on the energy and activity of the radiation source used, as well as the operating conditions of the detector [5]

Energy calibration require as many sources as possible to be able to account for the gamma detector energy range [6]. This is also needed to enhance the line of best fit for the calibration equation. It is difficult to get mono-energetic sources of wide energy range. Some standard poly-energetic sources with multiple energy lines have been used as standard sources in calibration [7]. Occasionally the 511 keV annihilation radiation peak can also be integrated as a source for energy calibration of gamma detectors [8]. This study seeks to make use of such gamma sources and radiation peaks to derive a more reliable calibration curve. The sources used are Cesium-137 (Cs-137), Cobalt-60 (Co-60) and Sodium-22 (Na-22) point sources. The decay schemes of the Cs-137, Co-60 and Na-22 sources are described in figure 1a, figure 1b. and figure 1c. respectively. The sources decay by either beta minus (β^-), beta plus (β^+), gamma emission or by electron capture.

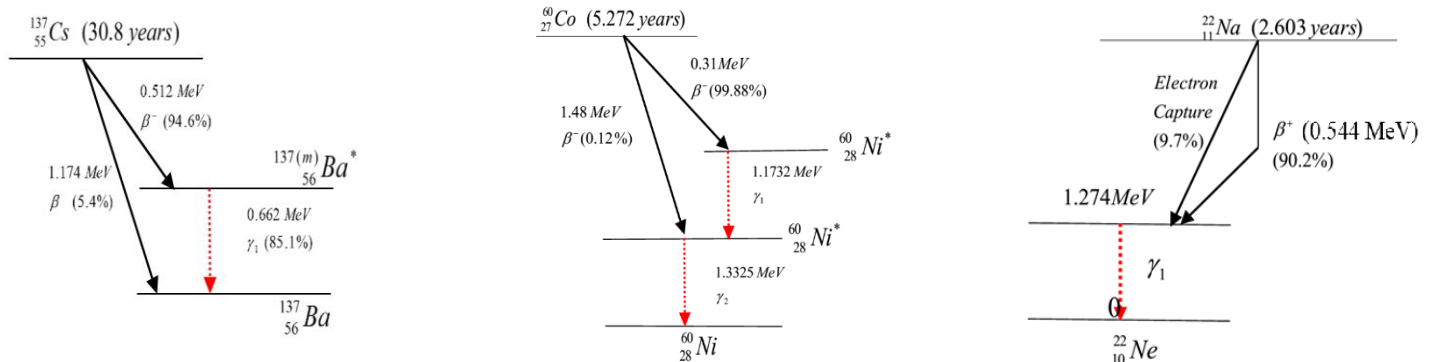


Figure 1. a. Cs-137 Decay Scheme b. Co-60 Decay Scheme c. Na-23 Decay Scheme

Cs-137 and Co-60 have been studied [9] using Monte Carlo method computational energy calibration in validating the experimental energy calibration data. A variety of Monte Carlo based techniques are available in literature [10; 11; 12; 13; 14; 15; 16; 17] to simulate the random processes that are probable in the gamma transport within the detector medium and derive detector response functions during computational energy calibration. Na-22 study is being added to the mix of sources and evaluated after simulations. The three (3) sources give five (5) prominent energy peaks in their gamma spectrum. Cs-137 gives a prominent peak at 661 keV, Co-60 also give two prominent peaks at 1173 keV and 1332 keV respectively, Na-22 source yields a 1274 keV peak and a 511 keV Annihilation radiation peak in its spectrum.

Computational means of energy calibration have been shown to compare similarly with those of experiment. In practice, gamma detectors like the NaI(Tl) are highly sensitive to gamma sources. As shown in the gamma spectroscopy setup in figure 2, the energy from the sources is deposited after interacting with the crystal volume of detectors and are converted to voltage pulses which are amplified and distributed into channels based on pulse counts through nuclear instrumentation. The counts eventually are read out as proportional factors (pulses) of the energy deposited in channels. This counts-channel output is then read out on the computer as gamma spectrum signatures of the gamma sources.

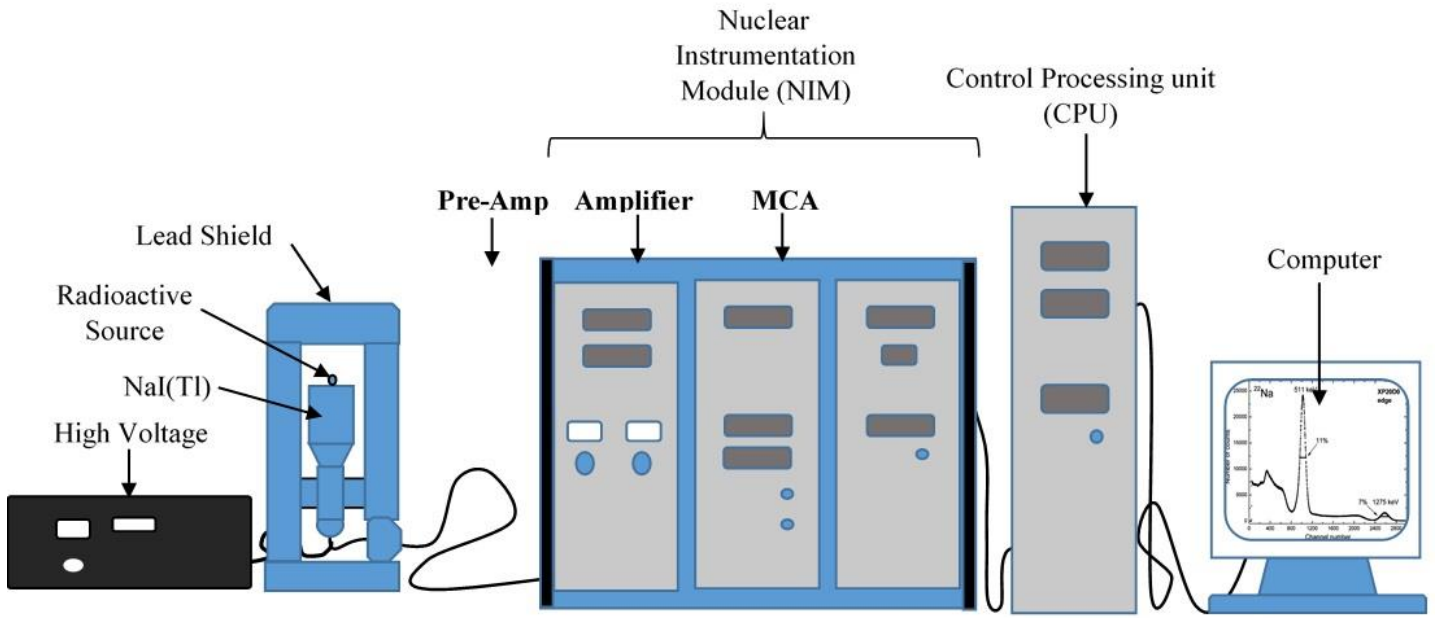


Figure 2. NaI(Tl) Detector System

2. Methods

2.1 Details of Experimental Setup

The experimental energy calibration process was carried out with 2 gamma sources of known activities, a 3"x3" NaI(Tl) scintillation detector, nuclear instrumentation setup and a computer readout monitor. The gamma sources considered were Cs-137 and Co-60 considered by [9]. The experimental process was described [9]. The spectrum from experimental procedure were compared with those from the MC. Simulations of the same calibration setup were carried out by the authors. The simulated spectra and response functions compared similarly with those in experiment. A calibration equation was derived based on the study which was expressed as

$$E(\text{keV}) = (0.667692 \cdot \text{channelnumber}) + 168.985 \quad (1)$$

In this study, the simulated sources were further increased to three, with Na-22, to further enhance the line of best fit on the calibration curve and hence remodify the calibration equation. The total energy points have been increased from 3 (Cs-137: 661 keV, Co-60: 1172 keV and 1332 keV) to 5 (with additional Na-22: 511 keV, 1274 keV).

2.2 Monte Carlo Method of Simulation

The Monte Carlo technique is an alternative to the analytical solution to the photon transport equation [18] since it can allow more complicated geometries and more numbers of photon particles. The MC. technique was

adopted the Fortran code called CalXTM written to simulate the random gamma photon transport based on the experimental setup, (i.e.) interaction with the detector crystal volume, the deposition of energy based on the interaction selected randomly, conversion to light (scintillation) and secondary electrons which lead to voltage-pulse production, amplification and binning of voltage pulse into channels in the multichannel analyzer (MCA) of the NIM. The MC. Algorithm uses pseudorandom uniform numbers [19] generated as a function in Fortran. The result of the MC technique is a count-channel graph or spectrum which is then converted to count-energy spectrum, for each gamma source using the energy calibration equation. This spectrum is read out on the monitor screen of the computer connected to the NIM. The NIM module comprises of the preamplifier(preamp), amplifier, analog-to-digital converter (ADC) and Multi-channel analyzers (MCA).

The common interaction types to consider in this study are photoelectric effect and Compton scattering (for energies lower than 1022 keV), and for energies higher, a pair-production (and pair annihilation) interaction is considered probable with the other two common interactions [20]. Energy-dependent cross sections from [21] were obtained for use in the code to sample for the interactions probability density functions (pdfs). For photoelectric effect (PE), gamma sources deposit energies (E) which are above the threshold frequency(ϕ) of ejecting electrons from the NaI(Tl) crystal according to the equation (2).

$$E = \phi + E_K \quad (2)$$

where E_K is the kinetic energy giving to the electron

For Compton scattering interaction(CMPT), the energetic photon from the gamma source interacts with tightly bound electrons and is scattered in random directions to interact through any type of interaction, while depositing part of its energy to the recoil electron. The governing equation is expressed in equation (3). This Compton event continues until the photon energy reaches a non-detectable threshold before it is assumed fully deposited.

$$E' = \frac{E}{1 + (E/m_0c^2) \cdot (1 - \cos \theta)} \quad (3)$$

where E' is the scattered photon energy, E is the incident photon energy, m_0c^2 is the rest energy of an electron and θ is the scattered angle of the gamma photon.

In the case of pair production (PP), however, gamma photons of energies above twice the rest energy of an electron (1022 keV) interact close to the nuclear field of atoms of the crystal and are converted to energized

electrons and positrons. These electrons and positrons keep scattering within the detector volume until it loses all kinetic energy. The positrons in the volume are eventually slowed down to rest and they annihilate with electrons of rest energies to form two pairs of photons with 511 keV energies each.

$$E = E_K + 1022keV \quad (4)$$

where E_K is the extra energy possessed by the gamma photon beyond the energy required to start a pair production interaction (1022 keV). These energies are either deposited or lost from the NaI crystal energy deposition in the NaI crystal volume.

The significance of Beta plus (β^+) decay is mostly felt during the pair annihilation event between rest electrons and rest positrons. Two 511 keV energy photons produced travel in opposite directions until they are absorbed, scattered or they escape the detector volume. Energetic positrons are transported in the detection medium and interactions which are possible occur (both collisional and radiational) such that continuous slowing down approximation range (CSDA) can be estimated from their collision and radiative stopping powers as given by the relations

$$R_{CSDA} = \int_0^{T_0} \left(\frac{dT}{\rho dx} \right)^{-1} dT \quad (5)$$

where T is the kinetic energy of the incoming charge particle (positron), ρ is the density of the (NaI(Tl)) detector medium and x is the mean free distance

The positrons are made to end their transport within an estimated maximum penetration depth (t_{max}) given with an expectation value of

$$< t > = -\frac{1}{N_0} \int_0^{\infty} t \cdot t_f(t) dt \quad (6)$$

The maximum penetration depth (t_{max}) is comparable to the CSDA range (R_{CSDA}) for low-Z materials but show wide variation in high-Z materials like NaI(Tl) crystal volume ($Z_{eff} = 53$ g/mol). The NaI(Tl) detector is said to have a CSDA range of 0.32459 g/cm² from interpolation of stopping power data at 544 keV of positron kinetic energy and a Z_{eff} of 3.67 g/cm³. The penetration depth is estimated at 1.2cm when we estimate that the mazimum

penetration depth equals the CSDA range (i.e. $t_{\max} = R_{\text{CSDA}}$). More rest electrons are generally produced within the detector volume after energy depositions within the $t_{\max}$ will come to rest and interact with rest positrons. If the β^+ decay is significantly high from the decay scheme, it has to be factored into the MC system based on sampling its branching ratio. In doing so, a significant single escape peak due to 511 keV positron and 511 keV electron pair annihilation event will be prominent. The deposited energy in the crystal volume subsequently releases photoelectrons with kinetic energies which lead to some scintillation on the crystals, and 85 more production of electrons in the dynode of the detector's photomultiplier tube (PMT). These electrons are then converted to signals that can be analyzed on PC monitor readout.

2.2.1 MC Algorithm for Solid Angle Determination

In the choice of which gamma (γ) and beta-plus (β^+) particles emitted by standard sources are received at the top surface area of the detector, the general random sampling technique for the solid angle estimated is

$$\Omega = 2\pi(1 - \cos(\theta)) \quad (7)$$

where θ ; $0^\circ \leq \theta \leq 180^\circ$, $\Rightarrow \theta = r.180^\circ$, $r : (0,1)$ interval

2.2.2 MC Algorithm for Emission Probabilities (Electron Capture (EC), Beta Plus(β^+), Beta Plus (β^-))

For the sources considered in this study most transitions in the decay scheme required EC, β^+ or β^- accompanied with gamma emission. Electrons and positrons lose significant energy through ionization and radiation, some of which contribute to characteristic x-ray peaks and bremsstrahlung effects. Sources with more β^+ emission probability are more likely to have their annihilation radiation lead to single-escape peaks due to 1022 keV annihilation energy which accumulate also for energy depositions

For the code;

For energies equal or below this cut off, photoelectric event is recorded. The following interaction events were probable above a desired cut-off energy. Many of these events lead eventually to an energy deposition of some sort.

(with less significant % β^+)

Photoelectric events: (#sampled energy depositions will be binned) (8)

Compton Scattered events: (#sampled energy depositions will be binned) (9)

Pair production/Annihilation:

$$E_k(\beta^+, \beta^-) = (\text{electron full kinetic energy depositions from Compton collision events}) \quad (10)$$

$$E_k(\beta^+, \beta^-) = (\text{positron full kinetic energy depositions from Compton collision events}) \quad (11)$$

(electron+ positron annihilation event, Single-Escape Peaks (SEP))

$$(\text{electron+ positron annihilation event, Double-Escape Peaks (SEP)}) \quad (12)$$

(significant % β^+)

Photoelectric events:

$$(\text{\#sampled energy depositions will be binned}) \quad (13)$$

Compton Scattered events:

(#sampled energy depositions will be binned)

$$E_k(\beta^+, \beta^-) = (\text{positron full kinetic energy depositions from Compton collision events})$$

$$(\text{Annihilation (positron rest energy + electron rest energy) leading to Single-Escape 511 keV Peaks (SEP)}) \quad (14)$$

Also, all energy deposited are sampled to check whether they fall with the region of maximum penetration for decaying positrons. If any PE or CMPT or PP interaction occurs in that region, it is assumed the resulting electrons ejected will come to rest and interact with trapped positron at rest and annihilate, leading to a significant 511 keV SEP

The structure of the CALX code is shown in figure (3), where the red arrow pointers help extract photons which emerge from a Compton event and are opportune to interact within the detector volume once more.

2.3 The Calibration Equation

The calibration equation used is expressed as [22]

$$E(\text{keV}) = I(\text{keV}) + G \cdot C(\text{Channels}) \quad (15)$$

where E is the energy obtained from the calibration curve, I and G are the intercept and gradient of the calibration line and C is the channel position

3. Results and Discussions

In the search for more reliable energy calibration curves, and energy calibration equation, the MC. algorithm has been enhanced to cater for the influence of standard sources with significant β^+ -emission probability. In this study, 10^6 gamma photons were emitted directly into the detector surface while the source was placed on the detector face. β^+ particles incident on the detector surface were also estimated using a solid angle evaluation. The

idealized Na-22 used is assumed to have comparable activity as the other sources i.e. the M.C. method will be subjected to the same calibration equation used on the other 2 sources.

3.1 Without β^+ consideration in Algorithm

Figure 4a shows the Na-22 spectrum without the MC Algorithm modification. Photoelectric absorption and Compton scattering event resulted more in the energy depositions used in generating the prominent 1274 keV photopeaks and of the Na-22 spectrum. Pair production and annihilation led to less prominent peaks despite its involvement as a competing interaction.

3.2 With β^+ consideration in Algorithm

Figure 4a shows Na-22 spectrum with MC. Algorithm modification. The 1274 keV peak in Figure 4b was not as prominent as that of figure 6 after the treatment of the code with the 1022 keV (β^+ - β^-) annihilation effect. The single escape peak of 511 keV as shown in figure 4b is reflective of the actual expected peak in Na-22 gamma spectrum in experiment. From the figures 4 above, the spectrum generated from figure 4a. indicates the energy buildup from deposition due to three significant interactions: Photoelectric, Compton and pair production without proper consideration of the abundance of β^+ particles emitted from the Na-22 source.

3.3 Comparison of M.C. generated Na-22 and Experimental Na-22 Spectrum

The MC. generated Na-22 spectrum was compared with those from experiment [23] and the response parameters was compared with those of [1]

From the spectrum as shown in figure 5b, a good similarity was found around the prominent peaks (1274 and 511 keV respectively). The background noise area of the experimental spectrum has been omitted in the M.C. spectrum of figure 5a because it has not been considered for an ideal situation. Also the characteristic x-ray peaks shown in the experimental spectrum was not accounted for since the energy cut-off was from 170 keV in the MC. simulated spectrum.

The FWHM and energy resolution for the MC. spectrum have been compared with those of [1] and the comparison is shown in table 1. About 8% error in the FWHM have been estimated for the 1274 keV peak while the energy resolution value from M.C. simulation varied with about 7% from experiment. For the 511 keV peak, the FWHM have been estimated with a value of 36 keV which is about 7% deviation from experimental values

of 39 keV. The energy resolution in the MC. simulation also showed a 7% deviation from experimental values with an MC. value of 7.05% as against 7.60% from experimental values. These deviations are minimal therefore similar agreement between the MC. simulation and experiment have been established

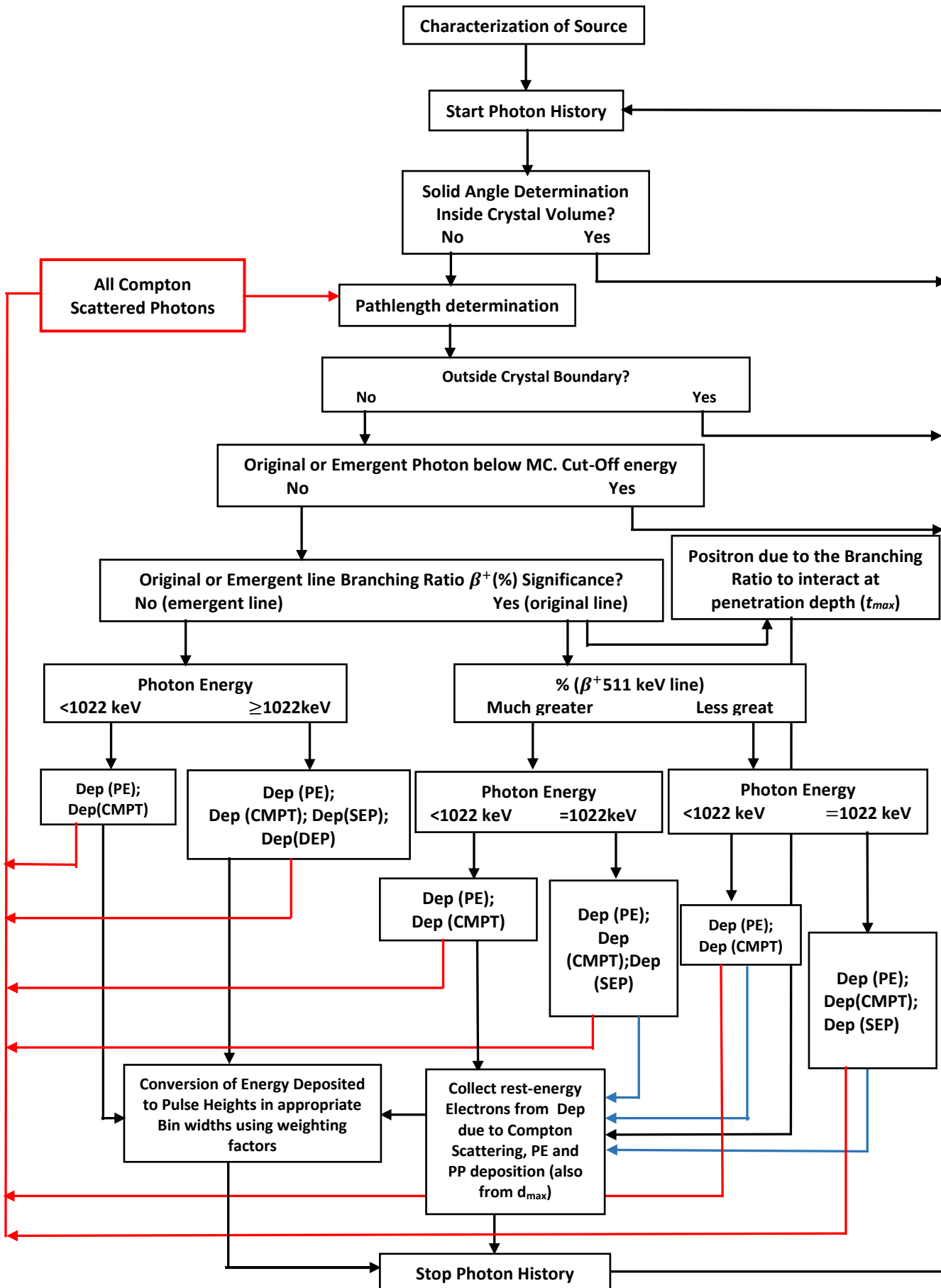
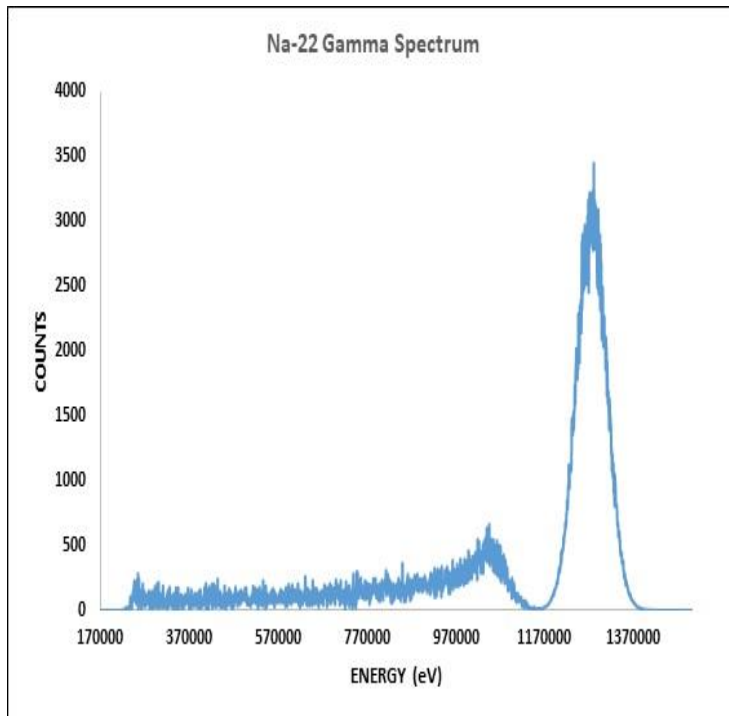
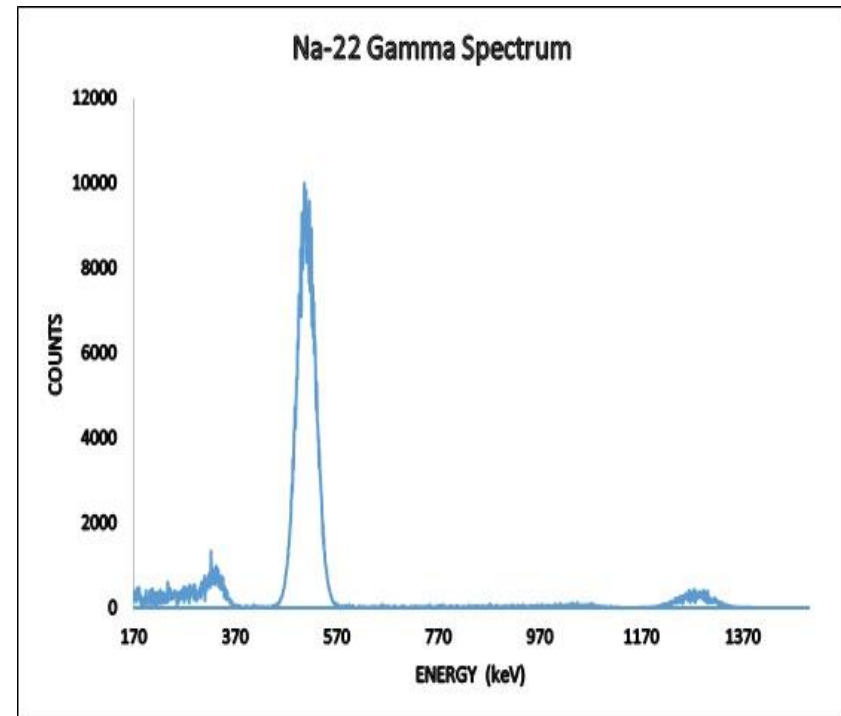


Figure 3. Structure of the CalX Code

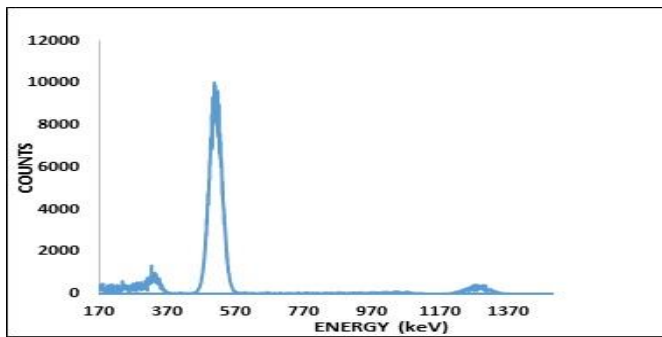


(a)

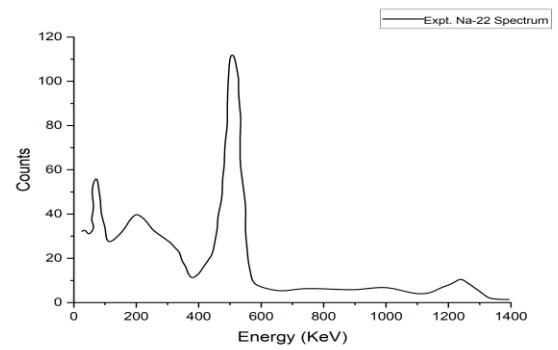


(b)

Figure 4. a. Na-22 spectrum without the MC Algorithm modification b. Na-22 spectrum with the MC Algorithm modification



a. MC. Simulated Na-22 Spectrum



b. Experimental Na-22 Spectrum [23]

Figure 5. A comparison of MC. Simulated Na-22 Spectrum (a.) with Experimental Na-22 Spectrum (b.) (Source: (b) from Vojtech Ullmann, 2020.)

Table 1. Detector Response Parameter Comparisons for the MC. Spectrum and the Experimental Spectrum

Na-22 $T_{1/2}=2.6$ y	Energy (keV)		FWHM(keV)	Detector Resolution(%)
	1274.54	M.C Simulation	47.00	3.74
		Akkurt <i>et al.</i> (2014)	51.00	4.00
	511	M.C Simulation	36.00	7.05
		Akkurt <i>et al.</i> (2014)	38.84	7.60

3.4 Modified Calibration Equation

The calibration curve now, as shown in figure 6, shows increased energy points (511 keV, 661 keV, 1172 keV, 1332 keV, 1274 keV) as compared with the one in the results from [9]. The new calibration line has been fitted so has to have a linear equation, with an r^2 statistical measure of 1, to describe the fit, between the channel to the energy response. The new equation is now expressed as

$$E(keV) = (0.6657 \cdot channelnumber) + 172.45 \quad (16)$$

4. Conclusions

In this research, a program named CALX was successfully written to compute the NaI(Tl) detector response parameters and calibration equation necessary for energy calibration process in NaI(Tl) detector facilities. These parameters were obtained for a simulated 3-inch diameter by 3-inch high NaI(Tl) detector crystal volume.

Energies from standard gamma photon sources, Cs-137 and Co-60, placed axially on the surface of the detector, were supplemented with an idealized Na-22 source energies to modify the calibration equation necessary to capture more energy points during energy calibration.

The MC spectrum plots for Cs-137 and Co-60 have shown agreements with experimental plots in an earlier paper [8] The addition of an idealized Na-22 source resulted in a different spectrum from the expected due to branching ratio factors which have not been fully incorporated in the PTRAN code. This had significant effect for Sources β^+ decay in its decay scheme. CalX code resulted in a far comparable spectrum when comparing the MC simulated spectrum with experimental plots for such kind of sources. Other minimal disagreements with

experimental spectrum were also obtained due to background noise, detection limits, cladding of detector material and other complications of the electronics from the Photomultiplier tubes to the MCA which have not been incorporated in the simulated NaI(Tl) detector algorithm.

Due to these results, Na-22 source was simulated and resulting spectrum gave a new calibration equation which gives a better linear proportionality of the fitting of channels to energy (in keV). Considering the results of this work, the outputs of CALX can be relied on for calculating response functions of unavailable gamma sources.

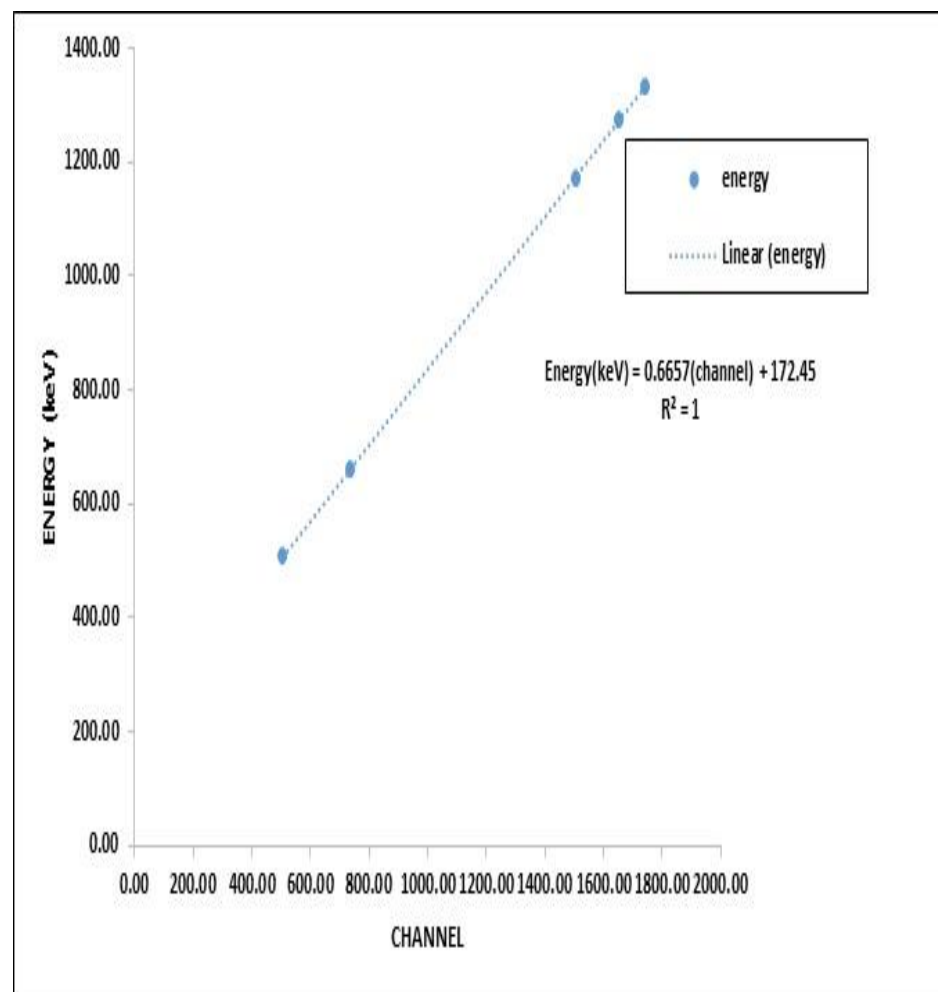


Figure 6. the calibration curve showing the estimated calibration equation for three standard gamma sources

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